

Solution outline

Main issues in the buggy code:

1. **Softmax is computed across the wrong axis**
It should normalize across classes for each example: `axis=1`, not `axis=0`.
2. **Softmax is numerically unstable**
Use `shifted = logits - logits.max(axis=1, keepdims=True)` before exponentiating.
3. **Cross-entropy gradient scaling is wrong**
Divide by batch size `n`, not by the number of classes.
4. **The second layer update uses gradient ascent**
`w2` and `b2` should be updated with `-=`, not `+=`.
5. **Normalization uses the full dataset**
That leaks validation information. Fit normalization on train only.
6. **Training order never changes**
Shuffle minibatches each epoch.